MPG No. 33.1.131 $\bar{4}2m$ (-4m2 setting) [Type I, tetragonal] [M tensor]

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} 0 & M_{xy} & 0 \\ M_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xy} = \mathbb{M}_{xy}^{(1)}$$

* Rank 2 tensor (a). * Rank 3 tensor (ss). * Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{yzx} & 0 & 0 \\ 0 & -M_{yzx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzx} = 3\mathbb{T}_{u}^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} M_{yzx} & 0 & 0 \\ 0 & M_{yzx} & 0 \\ 0 & 0 & M_{xyz} \end{bmatrix}$$

$$M_{yzx} = \mathbb{T}_{0}^{(1)} - \mathbb{T}_{u}^{(2)}$$
$$M_{xyz} = \mathbb{T}_{0}^{(1)} + 2\mathbb{T}_{u}^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & M_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & M_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & M_{zzxy} \\ 0 & 0 & 0 & 0 & M_{yzzx} & 0 \\ 0 & 0 & 0 & M_{yzzx} & 0 & 0 \\ M_{xxxy} & M_{xxxy} & M_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxy} = -\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)}$$
$$M_{zzxy} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)}$$
$$M_{yzzx} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & M_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & M_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & M_{zzxy} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -M_{xxxy} & -M_{xxxy} & -M_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxy} = \mathbb{M}_{xy}^{(3)} + 2\mathbb{T}_{z}^{\beta(1)}$$
$$M_{zzxy} = \mathbb{M}_{xy}^{(3)} - 4\mathbb{T}_{z}^{\beta(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} 0 & M_{yzzx} & 0 \\ M_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzzx} = 2\mathbb{M}^{(4)}_{xy}$$

* Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & M_{xxxy} \\ 0 & 0 & -M_{xxxy} \\ 0 & 0 & 0 \\ 0 & M_{yzzx} & 0 \\ -M_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxy} = -2\mathbb{M}^{(5)}_{xy} + \mathbb{T}^{\beta(2)}_{z}$$
$$M_{yzzx} = -\mathbb{M}^{(5)}_{xy} - \mathbb{T}^{\beta(2)}_{z}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{yzzx} & 0 \\ 0 & 0 & 0 & -M_{yzzx} & 0 & 0 \\ M_{xyxx} & -M_{xyxx} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzzx} = \mathbb{M}^{(6)}_{xy} + \mathbb{T}^{\beta(3)}_{z}$$
$$M_{xyxx} = -2\mathbb{M}^{(6)}_{xy} + \mathbb{T}^{\beta(3)}_{z}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & M_{xxxy} & 0 & 0 & M_{xxyx} \\ 0 & 0 & 0 & 0 & 0 & M_{xxyx} & 0 & 0 & M_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & M_{zzxy} & 0 & 0 & M_{zzxy} \\ 0 & 0 & 0 & 0 & M_{yzzx} & 0 & 0 & M_{yzxz} & 0 \\ 0 & 0 & 0 & M_{yzxz} & 0 & 0 & M_{yzzx} & 0 & 0 \\ M_{xyxx} & M_{xyxx} & M_{xyzz} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxy} = -\mathbb{M}^{\beta(1)}_{4z} + \mathbb{M}^{(1)}_{xy} + 2\mathbb{M}^{(2)}_{xy} + \mathbb{M}^{(3)}_{xy} - 2\mathbb{M}^{(5)}_{xy} + 2\mathbb{T}^{\beta(1)}_{z} + \mathbb{T}^{\beta(2)}_{z}$$
$$M_{xxyx} = -\mathbb{M}^{\beta(1)}_{4z} + \mathbb{M}^{(1)}_{xy} + 2\mathbb{M}^{(2)}_{xy} + \mathbb{M}^{(3)}_{xy} + 2\mathbb{M}^{(5)}_{xy} + 2\mathbb{T}^{\beta(1)}_{z} - \mathbb{T}^{\beta(2)}_{z}$$
$$M_{zzxy} = 2\mathbb{M}^{\beta(1)}_{4z} + \mathbb{M}^{(1)}_{xy} + \mathbb{M}^{(3)}_{xy} - 4\mathbb{T}^{\beta(1)}_{z}$$
$$M_{yzzx} = 2\mathbb{M}^{\beta(1)}_{4z} + \mathbb{M}^{(2)}_{xy} - \mathbb{M}^{(5)}_{xy} - \mathbb{T}^{\beta(2)}_{z}$$
$$M_{yzxz} = 2\mathbb{M}^{\beta(1)}_{4z} + \mathbb{M}^{(2)}_{xy} + \mathbb{M}^{(5)}_{xy} + \mathbb{T}^{\beta(2)}_{z}$$
$$M_{xyxx} = -\mathbb{M}^{\beta(1)}_{4z} + \mathbb{M}^{(1)}_{xy} + 2\mathbb{M}^{(2)}_{xy} - \mathbb{M}^{(3)}_{xy} - 2\mathbb{T}^{\beta(1)}_{z}$$
$$M_{xyzz} = 2\mathbb{M}^{\beta(1)}_{4z} + \mathbb{M}^{(1)}_{xy} - \mathbb{M}^{(3)}_{xy} + 4\mathbb{T}^{\beta(1)}_{z}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{yzzx} & 0 & 0 & M_{yzxz} & 0 \\ 0 & 0 & 0 & -M_{yzxz} & 0 & 0 & -M_{yzzx} & 0 & 0 \\ M_{xyxx} & -M_{xyxx} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzzx} = 2\mathbb{M}^{(4)}_{xy} + \mathbb{M}^{(6)}_{xy} + \mathbb{T}^{\beta(3)}_{z}$$
$$M_{yzxz} = -2\mathbb{M}^{(4)}_{xy} + \mathbb{M}^{(6)}_{xy} + \mathbb{T}^{\beta(3)}_{z}$$
$$M_{xyxx} = -2\mathbb{M}^{(6)}_{xy} + \mathbb{T}^{\beta(3)}_{z}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & M_{xxxy} & 0 \\ M_{xxxy} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & M_{zzxy} & 0 \\ M_{xxyx} & 0 & 0 \\ M_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & M_{xyyy} & 0 \\ 0 & 0 & -\frac{M_{xxxy}}{2} - \frac{M_{xxyx}}{2} + M_{xyyy} + M_{zzxy} \end{bmatrix}$$

$$M_{xxxy} = -\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} - 2\mathbb{M}_{xy}^{(5)}$$
$$M_{zzxy} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + \mathbb{M}_{xy}^{(3)}$$
$$M_{xxyx} = -\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_{xy}^{(5)}$$
$$M_{yzzx} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(5)}$$
$$M_{xyyy} = -\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(3)}$$